



Credit risk II.

Risk Management in Financial Institutions

Lecture 9

Béla ÖCSI

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Reminder: Credit Risk

- **Market risk** is the present and/or future **risk of losses** on on-balance sheet and off-balance sheet **positions** due to **changes in market prices** (changes in the price of bonds, securities, commodities, currencies or interest rates affecting the positions).
- **Credit risk** is the risk of loss to the profitability and capital position **arising from the non-performance (or nonperformance according to contractual terms) of counterparties**, i.e. the risk of **loss** resulting from partial or total failure to meet obligations (on or off balance sheet) to the financial institution.

Source : MNB ICAAP handbook

(<https://www.mnb.hu/letoltes/icaap-ilaap-bma-kezikonyv-2021januar.docx>)

Reminder: Main credit risk subtypes

- **credit default risk:** the credit risk associated with certain financial and ancillary financial services or investment services activities (e.g. **lending and borrowing**, financial leasing, guarantees and sureties, other banking commitments, debt purchase);
- **counterparty credit risk:** the potential loss arising from the **failure** of a counterparty to a given (typically **derivative**) transaction **to meet its contractual obligations** before the transaction is closed (final settlement of cash flows).
- **counterparty credit risk (issuer risk):** the credit risk to the issuers of debt securities and equity securities, which includes the risk that the value of the security will decline because of a deterioration in the issuer's credit quality;
- **settlement risk:** the risk that a settlement through a payment and clearing and securities settlement system will not be completed as expected (even though the financial institution party fulfils its contractual obligation, it does not receive the consideration for it).

Source: Recommendation No 14/2021 (IX.16.) of the Magyar Nemzeti Bank on the commitment, measurement, management and control of credit risk (<https://www.mnb.hu/letoltes/14-2021-hitelkocka-ajanlas.pdf>) and MNB ICAAP Handbook (counterparty risk)

Reminder: Credit risk concepts

- **Default event:** the counterparty to a financial contract (e.g. loan contract, security (bond), derivative) fails to perform as contractually agreed
 - Security, derivative: fails to pay exactly within the given time limit (e.g. fails to pay interest or principal on a given day or after a given day + 30 days grace period)
 - Credit agreement: 'looser' than this, see Basel definitions (e.g. 90 days without payment)
- **Default risk->probability of default (PD, %):**
 - a risk factor for credit risk
 - the probability that a default event will occur for a given counterparty/contract within a given time period (e.g. one year)
- **Loss given Default (LGD, %)=1-Recovery**
 - Another risk factor for credit risk
 - Amount/loss remaining after the cash inflows (Recovery) after the default event
 - Given as a % of EAD; often set as a fixed value (not a variable) for simplicity
- **Exposure (at Default, EAD):** exposure, the total amount of claims (expected to be) outstanding at the time of default
 - The third risk factor for credit risk
 - In some cases it is a "given" that does not need to be modelled, e.g. for loans, the outstanding principal
 - Derivatives - to be modelled - positive expected market value at default
- **Expected Credit Loss (ECL or EL)=PD*LGD*EAD**

**FTX Group Companies Commence Voluntary Chapter 11 Proceedings in the United States
Begin Orderly Process to Review and Monetize Assets for Benefit of Global Stakeholders
John J. Ray III Appointed Chief Executive Officer; Sam Bankman-Fried Resigns**

FTX Trading Ltd. (d.b.a. FTX.com), announced today that it, West Realm Shires Services Inc. (d.b.a. FTX US), Alameda Research Ltd. and approximately 130 additional affiliated companies (together, the “FTX Group”), have commenced voluntary proceedings under Chapter 11 of the United States Bankruptcy Code in the District of Delaware in order to begin an orderly process to review and monetize assets for the benefit of all global stakeholders.

John J. Ray III has been appointed Chief Executive Officer of the FTX Group. Sam Bankman-Fried has resigned his role as Chief Executive Officer and will remain to assist in an orderly transition. Many employees of the FTX Group in various countries are expected to continue with the FTX Group and assist Mr. Ray and independent professionals in its operations during the Chapter 11 proceedings.

“The immediate relief of Chapter 11 is appropriate to provide the FTX Group the opportunity to assess its situation and develop a process to maximize recoveries for stakeholders,” said Mr. Ray. “The FTX Group has valuable assets that can only be effectively administered in an organized, joint process. I want to ensure every employee, customer, creditor, contract party, stockholder, investor, governmental authority and other stakeholder that we are going to conduct this effort with diligence, thoroughness and transparency. Stakeholders should understand that events have been fast-moving and the new team is engaged only recently. Stakeholders should review the materials filed on the docket of the proceedings over days for more information.”

Excluded Subsidiaries

The following subsidiaries are not included in the Chapter 11 proceedings: LedgerX | Digital Markets Ltd., FTX Australia Pty Ltd. and FTX Express Pay Ltd..

Default – how it works for a company, and how a bank can be saved in reality

Press Release

FDIC Acts to Protect All Depositors of the former Silicon Valley Bank, Santa Clara, California

Monday, March 13, 2023

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Media Contact:

 mediarequests@fdic.gov



About the FDIC:

The Federal Deposit Insurance Corporation (FDIC) is an independent agency created by the Congress to maintain stability and public confidence in the nation's financial system. The FDIC insures deposits; examines and supervises financial institutions for safety, soundness, and consumer protection; makes large and complex financial institutions resolvable; and manages receiverships.

For Release

WASHINGTON – The Federal Deposit Insurance Corporation (FDIC) today transferred all deposits—both insured and uninsured—and substantially all assets of the former Silicon Valley Bank of Santa Clara, California, to a newly created, full-service FDIC-operated ‘bridge bank’ in an action designed to protect all depositors of Silicon Valley Bank.

Depositors will have full access to their money beginning this morning, when Silicon Valley Bridge Bank, N.A., the bridge bank, opens and resumes normal banking hours and activities, including online banking. Depositors and borrowers will automatically become customers of Silicon Valley Bridge Bank, N.A. and will have customer service and access to their funds by ATM, debit cards, and writing checks in the same manner as before. Silicon Valley Bank’s official checks will continue to clear. Loan customers should continue making loan payments as usual.

Silicon Valley Bank was closed by the California Department of Financial Protection and Innovation on Friday, March 10, 2023, and the FDIC was appointed receiver.

The transfer of all the deposits was completed under the [systemic risk exception approved yesterday](#). All depositors of the institution will be made whole. No losses associated with the resolution of Silicon Valley Bank will be borne by taxpayers. Shareholders and certain unsecured debt holders will not be protected. Senior management has also been removed. Any losses to the Deposit Insurance Fund to support uninsured depositors will be recovered by a special assessment on banks, as required by law.

The receiver for Silicon Valley Bank has also transferred all Qualified Financial Contracts (as defined in 12 USC 1821(e)) of the failed bank to the bridge bank.

These actions will protect depositors and preserve the value of the assets and operations of Silicon Valley Bank, which may improve recoveries for creditors and the DIF.

A bridge bank is a chartered national bank that operates under a board appointed by the FDIC. It assumes the deposits and certain other liabilities and purchases certain assets of a failed bank. The bridge bank structure is designed to “bridge” the gap between the failure of a bank and the time when the FDIC can stabilize the institution and implement an orderly resolution.



Estimating Default Probabilities

- Alternatives:
 - Use historical data
 - Use credit spreads
 - Use Merton's model

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Merton's Model (Section 19.8, pages 419-422)

- Merton's model regards the equity as an option on the assets of the firm
- In a simple situation the equity value is

$$\max(V_T - D, 0)$$

where V_T is the value of the firm and D is the debt repayment required

Metron model:

Risky bond = $PV(D)$ - put option

Share of leveraged company = call option

where the underlying is the company's asset value (A or V)

the strike price is the face value of the loan (D)

the maturity is the maturity of the loan (T)

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Equity vs. Assets

An **option pricing** model enables the value of the firm's equity today, E_0 , to be related to the value of its assets today, V_0 , and the volatility of its assets, σ_V

$$E_0 = V_0 N(d_1) - De^{-rT} N(d_2) \quad \text{(1) Black-Scholes formula}$$

where

$$d_1 = \frac{\ln(V_0/D) + (r + \sigma_V^2/2)T}{\sigma_V \sqrt{T}} ; \quad d_2 = d_1 - \sigma_V \sqrt{T}$$

Volatilities

(2) Ito-Lemma

$$\sigma_E E_0 = \frac{\partial E}{\partial V} \sigma_V V_0 = N(d_1) \sigma_V V_0$$

This equation together with the option pricing relationship enables V_0 and σ_V to be determined from E_0 and σ_E

Default probability

From (1) and (2) we obtain

DD, Distance to Default

$$DD = \frac{\ln(A_0 / D) + (\mu - \sigma_A^2 / 2) \cdot T}{\sigma_A \cdot \sqrt{T}}$$

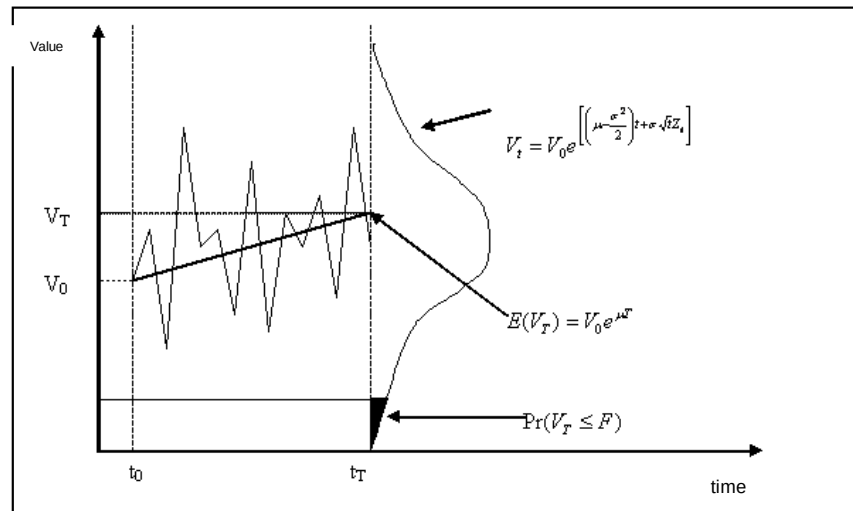
EDF(= Default probability=PD)

EDF = expected default frequency

EDF=1-N(DD)

in a risk-free world:

$$\text{EDF} = N(-d_2)$$



Example

- A company's equity is \$3 million and the volatility of the equity is 80%
- The risk-free rate is 5%, the debt is \$10 million and time to debt maturity is 1 year
- Solving the two equations yields $V_0=12.40$ and $\sigma_v=21.23\%$

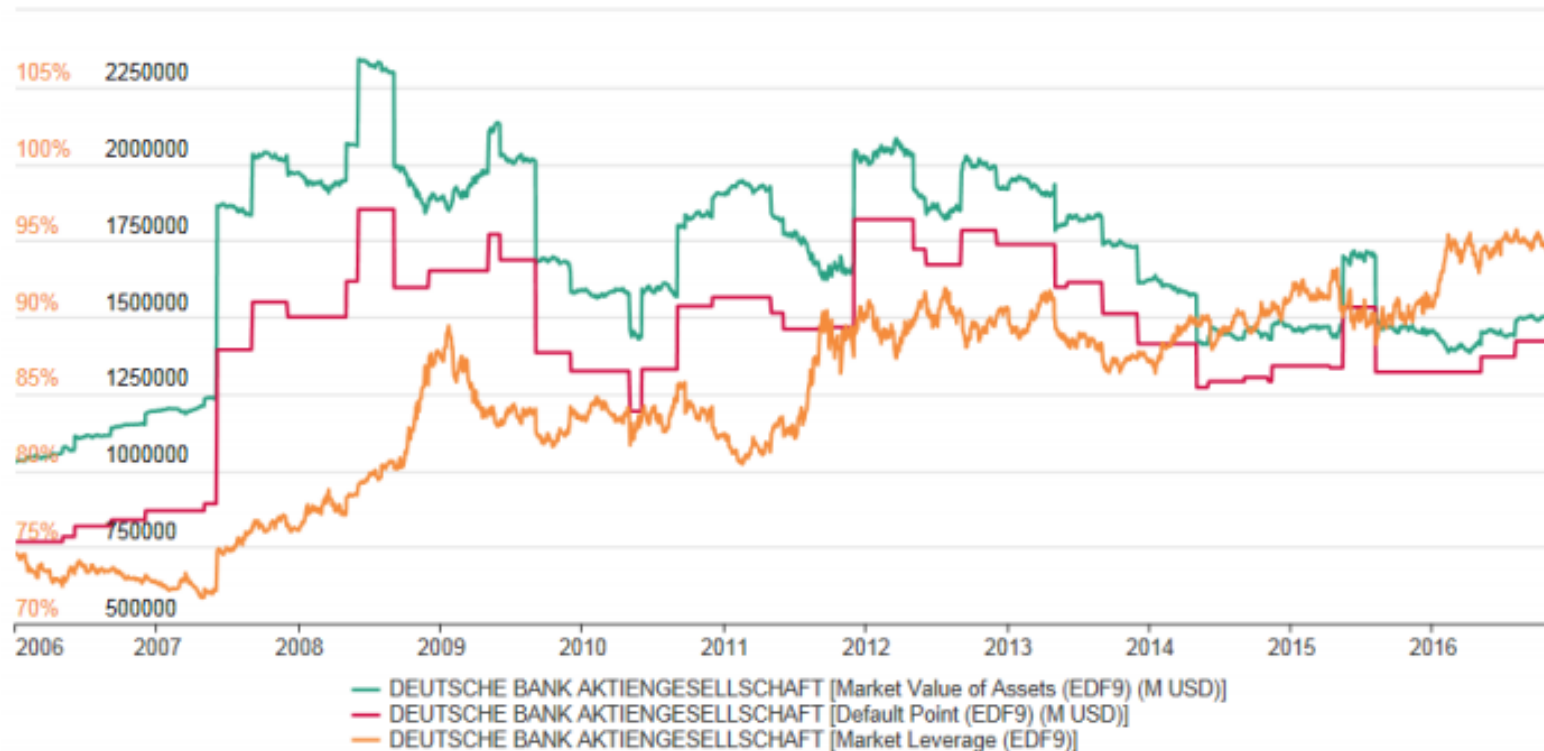
Example continued

- The probability of default is $N(-d_2)$ or 12.7%
- The market value of the debt is 9.40
- The present value of the promised payment is 9.51
- The expected loss is about 1.2%
- The recovery rate is 91%

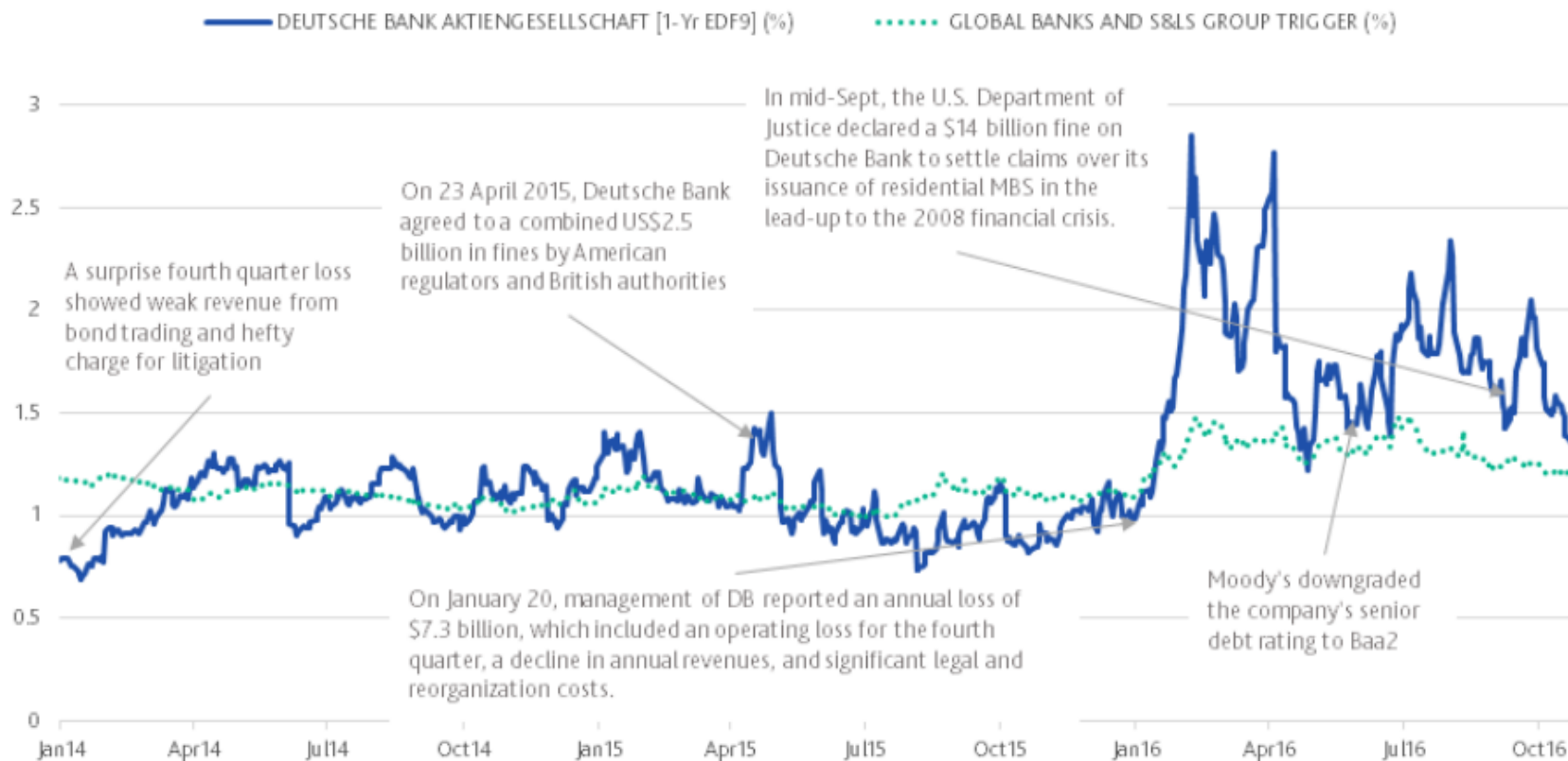
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Example: Deutsche Bank - asset value

FIGURE 6: Deutsche Bank's Market Leverage, Market Value of Assets, and Default Point



Example: Deutsche Bank EDF



<http://www.zerohedge.com/news/2016-10-28/moodys-warns-deutsche-bank-dangerously-close-falling-below-its-default-point>

Credit VaR

- Can be defined analogously to Market Risk VaR
- A one year credit VaR with a 99.9% confidence is the loss level that we are 99.9% confident will not be exceeded over one year
- Creditmetrics/KMV
- CreditRisk+

Credit VaR methodologies

- Structural/„Mark-to-Market” models
 - There is an economic model behind default (Merton)
 - Simpler normality assumptions: Vasicek ((IRB) model)
 - Full rating migration matrix estimate: CreditMetrics
- „Default mode”/”Actuary” models
 - The cause of default is not observable, it is like a damage event
 - Recursive summation of elementary distributions
 - Requires PD standard deviation in addition to PDs: CreditRisk+

Credit VaR methodologies

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 - Requires PD standard deviation in addition to PDs: **CreditRisk+**
- Combination of the two models
 - Simpler normality assumptions: **Vasicek**
 - Diversified portfolio, single factor model: Basel 2 **IRB**

Rating Transitions

- One year rating transition probabilities are published by rating agencies.
- If we assume that the rating transition in one period is independent of that in other periods we can calculate the rating transition for any period (see Appendix J and software)
- The “ratings momentum” phenomenon means that the independence assumption is not perfectly correct

Cumulative Average Default Rates % (1970-2013, Moody's) Table 19.1, page 402

	Time (years)						
	1	2	3	4	5	7	10
Aaa	0.000	0.013	0.013	0.037	0.104	0.241	0.489
Aa	0.022	0.068	0.136	0.260	0.410	0.682	1.017
A	0.062	0.199	0.434	0.679	0.958	1.615	2.759
Baa	0.174	0.504	0.906	1.373	1.862	2.872	4.623
Ba	1.110	3.071	5.371	7.839	10.065	13.911	19.323
B	3.904	9.274	14.723	19.509	23.869	31.774	40.560
Caa	15.894	27.003	35.800	42.796	48.828	56.878	66.212

One-Year Rating Transition Matrix (% probability, Moody's 1970-2013)

Table 21.1 page 446

Initial Rating	Rating at year end								
	Aaa	Aa	A	Baa	Ba	B	Caa	Ca-C	Default
Aaa	90.65%	8.67%	0.65%	0.00%	0.03%	0.00%	0.00%	0.00%	0.00%
Aa	0.95%	89.44%	8.95%	0.54%	0.07%	0.02%	0.01%	0.00%	0.02%
A	0.06%	2.56%	90.73%	5.86%	0.58%	0.11%	0.03%	0.00%	0.06%
Baa	0.04%	0.18%	4.20%	90.27%	4.17%	0.78%	0.16%	0.02%	0.18%
Ba	0.01%	0.06%	0.37%	6.17%	83.45%	8.06%	0.65%	0.07%	1.16%
B	0.01%	0.03%	0.12%	0.34%	5.08%	82.90%	6.69%	0.65%	4.18%
Caa	0.00%	0.02%	0.02%	0.12%	0.42%	9.74%	71.07%	4.06%	14.55%
Ca-C	0.00%	0.00%	0.07%	0.00%	0.43%	2.37%	10.59%	42.36%	44.20%
Default	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%

Five-Year Rating Transition Matrix

(calculated from one-year transitions)

Table 21.2 page 447

Initial Rating	Rating at end								
	Aaa	Aa	A	Baa	Ba	B	Caa	Ca-C	Default
Aaa	61.83%	28.85%	7.97%	1.02%	0.23%	0.06%	0.01%	0.00%	0.04%
Aa	3.19%	59.53%	30.10%	5.72%	0.90%	0.28%	0.08%	0.01%	0.19%
A	0.40%	8.67%	65.07%	20.36%	3.49%	1.12%	0.26%	0.04%	0.59%
Baa	0.18%	1.44%	14.58%	63.62%	12.63%	4.57%	0.98%	0.12%	1.87%
Ba	0.06%	0.39%	3.01%	18.39%	44.60%	20.67%	4.09%	0.46%	8.32%
B	0.04%	0.16%	0.71%	3.10%	12.88%	44.99%	12.85%	1.49%	23.79%
Caa	0.01%	0.07%	0.21%	0.83%	3.42%	18.63%	22.09%	2.75%	52.00%
Ca-C	0.00%	0.02%	0.14%	0.29%	1.27%	5.61%	7.16%	2.09%	83.40%
Default	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%

Rating change&default – an example

RESUPD — RATING_ACTION — 20 Mar, 2023 | 17:09 — Canada, United States of America, Latin America, EMEA, APAC, APAC

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Overview

- On March 19, 2023, Credit Suisse and UBS entered into a merger agreement following the intervention of the Swiss Federal Department of Finance, the Swiss National Bank (SNB), and the Swiss Financial Market Supervisory Authority (FINMA). UBS will be the surviving entity when the merger closes.
- In our view, Credit Suisse's acquisition by the stronger UBS group will stabilize its franchise and its funding and liquidity, while strengthening its governance and risk management standards.
- SNB has also announced that substantial liquidity support would be available, in addition to the Swiss franc (CHF) 50 billion facilities already provided to Credit Suisse.
- Because FINMA has determined that approximately CHF16 billion of additional Tier 1 (AT1) capital issued by Credit Suisse Group AG will be written down to zero, we have downgraded these AT1 instruments to 'C'; we expect them to default in the near term.
- We placed our 'A-/A-2' long- and short-term issuer credit ratings on the group's core operating subsidiaries, including principal operating bank Credit Suisse AG, on CreditWatch positive. We also placed our 'BBB-' long-term issuer credit rating on



Sector

Financial Institutions,
Banks, Non-banks

Tags

Americas, Latin America,
APAC, EMEA

Topic

Global Banking Risk

Primary Credit Analyst:

Anna Lozmann

Secondary Contacts:

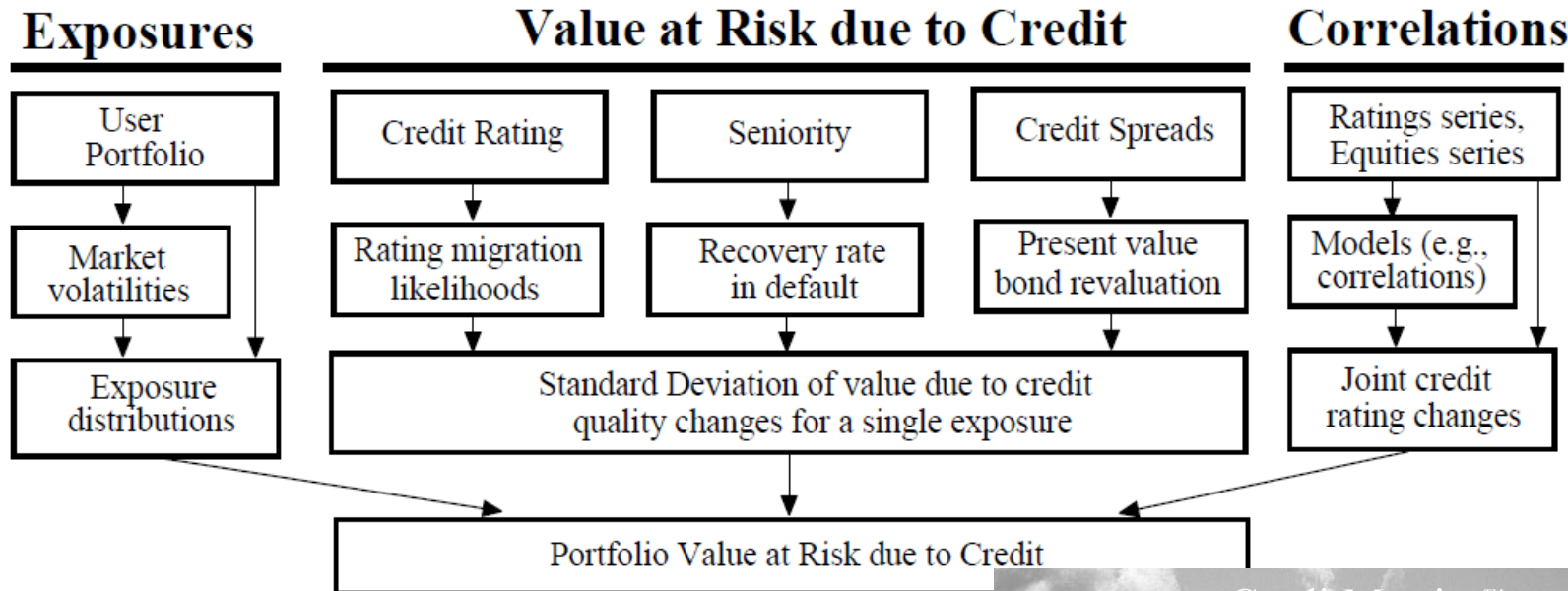
Benjamin Heinrich, Markus
W Schmaus

Ratings List

Downgraded

	To	From
Credit Suisse Group AG		
Junior Subordinated	C	B
Junior Subordinated	C	B+

CreditMetrics overview



New York
April 2, 1997

CreditMetrics™ – Technical Document

The benchmark for understanding credit risk

- A value-at-risk (VaR) framework applicable to *all institutions worldwide that carry credit risk in the course of their business.*
- A full portfolio view addressing credit event correlations which can identify the costs of over concentration and benefits of diversification in a mark-to-market framework.
- Results that drive: *investment decisions, risk-mitigating actions, consistent risk-based credit limits, and rational risk-based capital allocations.*

J.P. Morgan

Co-sponsors:

Bank of America

Bank of Montreal

BZW

Deutsche Morgan Grenfell

KMV Corporation

Swiss Bank Corporation

Union Bank of Switzerland

This *Technical Document* describes CreditMetrics™, a framework for quantifying credit risk in portfolios of traditional credit products (loans, commitments to lend, financial letters of credit), fixed income instruments, and market-driven instruments subject to counterparty default (swaps, forwards, etc.). This is the first edition of what we intend will be an ongoing refinement of credit risk methodologies.

Just as we have done with RiskMetrics™, we are making our methodology and data available for three reasons:

1. We are interested in promoting greater transparency of credit risk. Transparency is the key to effective management.
2. Our aim is to establish a benchmark for credit risk measurement. The absence of a common point of reference for credit risk makes it difficult to compare different approaches to and measures of credit risk. Risks are comparable only when they are measured with the same yardstick.
3. We intend to provide our clients with sound advice, including advice on managing their credit risk. We describe the CreditMetrics™ methodology as an aid to clients in understanding and evaluating that advice.

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CreditMetrics: components

Example: 1 bond rated BBB

Probability of credit rating migrations in one year for a BBB

Year-end rating	Probability (%)
AAA	0.02
AA	0.33
A	5.95
BBB	86.93
BB	5.30
B	1.17
CCC	0.12
Default	0.18

Calculation of year-end values after credit rating migration from BBB (\$)

Rating	Coupon	Forward Value	Total Value
AAA	6.00	103.37	109.37
AA	6.00	103.10	109.19
A	6.00	102.66	108.66
BBB	6.00	101.55	107.55
BB	6.00	96.02	102.02
B	6.00	92.10	98.10
CCC	6.00	77.64	83.64
Default	–	51.13	51.13

Credit VaR (1 year,
99,82%*)=56,42
USD
(=107,55-51,13)

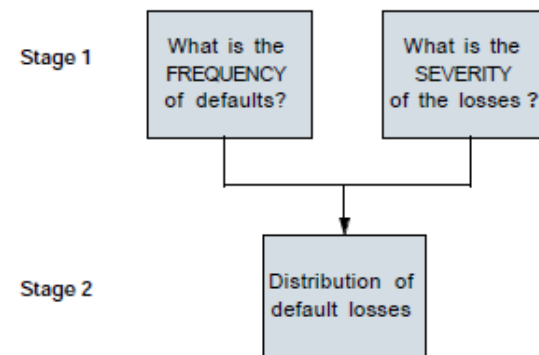
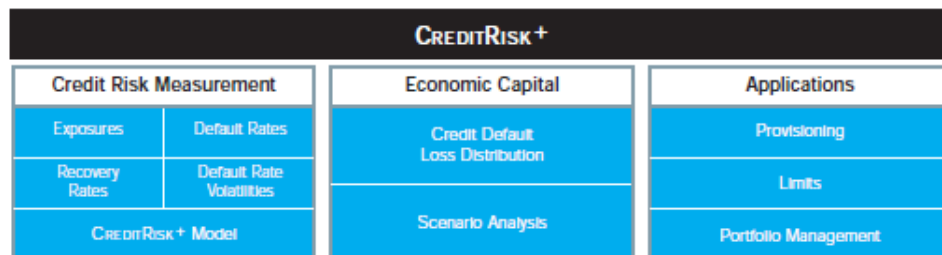
(*A confidence level of 99.82% is not only a "comfortable" choice in this example, but for a real BBB-rated bank it would be the "appropriate" level for the rating, since the probability of a loss resulting in bankruptcy, i.e. a loss greater than the VaR, is then 100-99.82=0.18% per year, which is exactly the 1-year PD of the BBB category)

Year-end rating	Value (\$)	Probability (%)
AAA	109.37	0.02
AA	109.19	0.33
A	108.66	5.95
BBB	107.55	86.93
BB	102.02	5.30
B	98.10	1.17
CCC	83.64	0.12
Default	51.13	0.18

Credit VaR methodologies

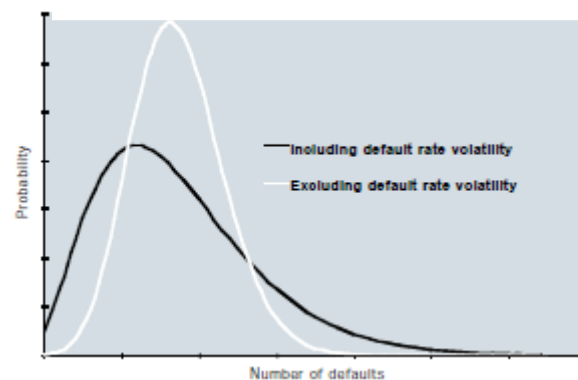
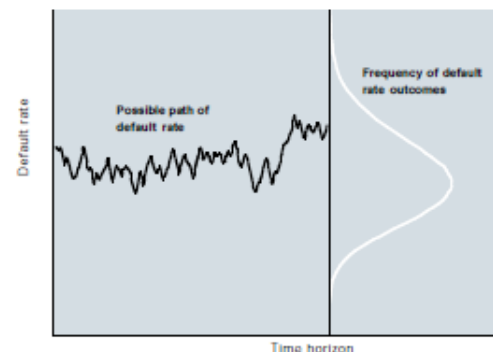
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CreditRisk+ (1)



Default distribution:

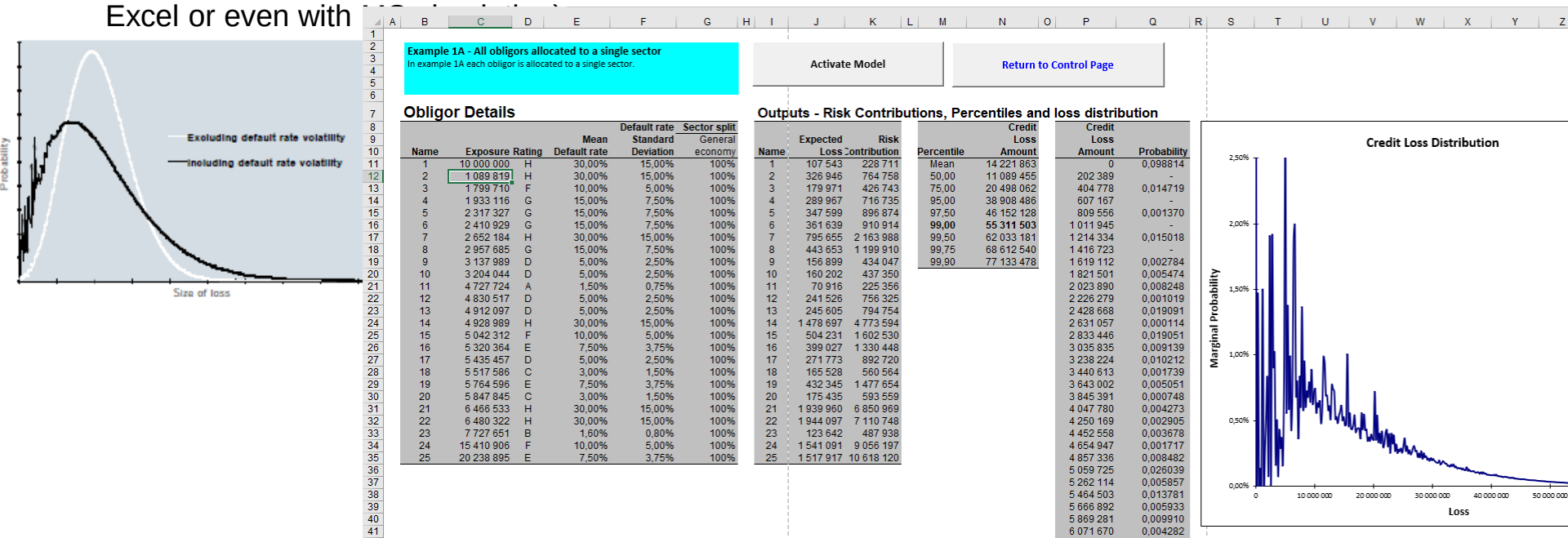
- If default rate is assumed to be known (no PD volatility) we can use a Poisson distribution to determine the probability of m defaults (see equation 21.3)
- If we assume the default rate has a gamma distribution (there is PD volatility) then the probability of m defaults has a negative binomial distribution (see equation 21.4 and Table 21.4)



CreditRisk+ (2)

Loss distribution

- The distribution of losses differs from the distribution of default events because the amount lost in a given default depends on the exposure to the individual obligors
- Unlike the variation of default probability between obligors, which does not influence the distribution of the total number of defaults, the variation in exposure magnitude results in a loss distribution that is **not Poisson in general**.
- However, it is possible to describe the overall distribution of losses because its probability generating function has a simple closed form amenable to computation
- The exposures are adjusted by anticipated recovery rates in order to calculate the loss in a given default.
- The exposures, net of the above recovery adjustment, are divided into **bands of exposure** with the level of exposure in each band being approximated by a common average.
- The CR+ Model calculates the probability that **a loss of a certain multiple of the chosen unit of exposure** will occur. This allows a full loss distribution to be generated (can be calculated with a macro in Excel or even with



Vasicek's Model (Equation 21.1, page 448)

- For a large portfolio of loans, each of which has a probability of PD of defaulting by time T the worst case default rate that will not be exceeded at the $X\%$ confidence level is

$$\text{WCDR} = N \left\{ \frac{N^{-1}[\text{PD}] + \sqrt{\rho} N^{-1}(X)}{\sqrt{1-\rho}} \right\}$$

where ρ is the Gaussian copula correlation

$$\text{VaR} = \text{WCDR} \times \text{EAD} \times \text{LGD}$$

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TANÁCSADÁS ÉS OKTATÁS



**Thank you for your
attention!**

